南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

考试科目： 高等数学(下) A **开课单位：** 数 学 系

考试时长： 120 分钟 **命题教师：** ______

题 号	1	2	3	4	5	6	7	8	9
分 值	15 分	15 分	10 分	10 分	10 分	10 分	10 分	10 分	10 分

本试卷共 9 道大题，满分 100 分. (考试结束后请将试卷、答题本、草稿纸一起交给监考老师)

注意：本试卷里的中文为直译（即完全按英文字面意思直接翻译），所有数学词汇的定义请参照教材（Thomas' Calculus，13th Edition）中的定义。如果其中有些数学词汇的定义不同于中文书籍（比方说同济大学的高等数学教材）里的定义，以教材（Thomas' Calculus，13th Edition）中的定义为准。

1. (15pts) **Multiple Choice Questions:** (only one correct answer for each of the following questions.)

   (1) If f is differentiable, and $z = z(x, y)$ is determined by $f(x - az, y - bz) = 0$, then $a\frac{\partial z}{\partial x} + b\frac{\partial z}{\partial y} =$

   (A) 1. (B) -1. (C) a (D) b.

   (2) Let $a_n > 0$ for all n. Which of the following statements must be **true** ?

   (A) If $\lim\limits_{n\to\infty} na_n = 0$, then the series $\sum\limits_{n=1}^{\infty} a_n$ converges.

   (B) If $\lim\limits_{n\to\infty} na_n = l$ and $l \neq 0$, then the series $\sum\limits_{n=1}^{\infty} a_n$ converges.

   (C) If $\lim\limits_{n\to\infty} na_n = l$ and $l \neq 0$, then the series $\sum\limits_{n=1}^{\infty} a_n$ diverges.

   (D) None of the above statements is correct.

   (3) Identify the surface of $2x^2 + y^2 = z^2$.

   (A) Hyperboloid of two sheets. (B) Elliptical Cone.

   (C) Hyperboloid of one sheet. (D) Elliptical paraboloid.

   (4) If $f(x, y) = \varphi(x + y) + \varphi(x - y) + \int_{x-y}^{x+y} \psi(t)dt$, where φ and ψ are twice differentiable functions, then

   (A) $\dfrac{\partial^2 f}{\partial x \partial y} = \dfrac{\partial^2 f}{\partial x^2}$. (B) $\dfrac{\partial^2 f}{\partial x \partial y} = \dfrac{\partial^2 f}{\partial y^2}$.

   (C) $\dfrac{\partial^2 f}{\partial x^2} = -\dfrac{\partial^2 f}{\partial y^2}$. (D) $\dfrac{\partial^2 f}{\partial x^2} = \dfrac{\partial^2 f}{\partial y^2}$.

   (5) $\lim\limits_{(x,y)\to(0,0)} (1 + xy)^{\frac{1}{x^2+y^2}} =$

   (A) 0 . (B) 1 .

   (C) e . (D) does not exist.

2. (15 pts) Fill in the blanks.

   (1) If $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are unit vectors and $\mathbf{a}+\mathbf{b}+\mathbf{c}=\mathbf{0}$, then $\mathbf{a}\cdot\mathbf{b}+\mathbf{b}\cdot\mathbf{c}+\mathbf{c}\cdot\mathbf{a}=$ _______.

   (2) If the vector $\mathbf{c}$ is perpendicular to $\mathbf{a}=\langle \mathbf{1},\mathbf{2},\mathbf{1}\rangle$ and $\mathbf{b}=\langle -\mathbf{1},\mathbf{1},\mathbf{1}\rangle$ and $\mathbf{c}\cdot(\mathbf{i}-\mathbf{2j}+\mathbf{k})=\mathbf{16}$, then $\mathbf{c}=$ _______ .

   (3) If $\displaystyle\sum_{n=2}^{\infty}\left(\tan\frac{1}{n}-k\ln\left(1-\frac{1}{n}\right)\right)$ converges, then $k=$ _______ .

   (4) The maximum curvature κ of function $y(x)=\sin x$ is _______ .

   (5) If $(z+y)^x=xy$, then $\frac{\partial z}{\partial x}(1,2)=$ _______.

3. (10 pts) Given a cardioid $r=a(1+\cos\theta), a>0$ and a circle $r=a$.

   (1) Find the area of the region that lies inside the cardioid and outside the circle.

   (2) Find the area of the region that lies inside the cardioid and inside the circle.

4. (10 pts) Assume $\mathbf{r}(t)=f(t)\,\mathbf{i}+g(t)\,\mathbf{j}+h(t)\mathbf{k}$, where

$$f(t)=\int_0^t \cos(x^2)\,dx, \quad g(t)=-t\cos t, \quad h(t)=\sum_{n=1}^{\infty}\frac{t^n}{n}.$$

   Calculate $\mathbf{r}'(0)$.

5. (10 pts) Let $f(x,y)=\begin{cases} y\sin\frac{1}{x^2+y^2}, & (x,y)\neq(0,0); \\ 0, & (x,y)=(0,0). \end{cases}$

   (1) Is $f(x,y)$ is continuous at $(0,0)$?

   (2) Find $f_x(0,0)$ and $f_y(0,0)$, if they exist.

6. (10 pts) Find the limit, if it exists, or show that the limit does not exist.

   (1) $\displaystyle\lim_{(x,y)\to(0,0)}\frac{xy}{\sqrt{x^2+y^2}}$.

   (2) $\displaystyle\lim_{(x,y)\to(0,0)}\frac{xy^3+2x^2y^4}{x^2+y^6}$.

7. (10 pts) For the power series $f(x)=\displaystyle\sum_{n=1}^{\infty}\frac{n+2}{n(n+1)}x^n$,

   (1) For what values of x does the power series converge ?

   (2) Find the sum of the series within the interval of convergence.

8. (10 pts) Determine if the series, $\displaystyle\sum_{n=2}^{\infty}\frac{(-1)^{n+1}}{n^p(\ln n)^2}\,(p>0)$, converges absolutely, or converges conditionally, or diverges. Give reasons for your answer.

9. (10 pts) Find $\displaystyle\lim_{n\to\infty}\left((n^2-n)e^{\frac{1}{n}}-\sqrt{n^4+1}\right)$.

一、　（15分）**单项选择题:**

(1) 设二元函数 f 可微，且 $z=z(x,y)$ 由方程 $f(x-az,y-bz)=0$ 确定，则 $a\frac{\partial z}{\partial x}+b\frac{\partial z}{\partial y}=$

(A) 1.　　(B) -1.　　(C) a　　(D) b.

(2) 已知对任意 $n>0$ 都满足 $a_n>0$，则下列哪一个结论一定成立?

(A) 若 $\lim\limits_{n\to\infty} na_n=0$, 则级数 $\sum\limits_{n=1}^{\infty} a_n$ 一定收敛.

(B) 若 $\lim\limits_{n\to\infty} na_n=l$，这里 $l\neq 0$, 则级数 $\sum\limits_{n=1}^{\infty} a_n$ 一定收敛.

(C) 若 $\lim\limits_{n\to\infty} na_n=l$，这里 $l\neq 0$, 则级数 $\sum\limits_{n=1}^{\infty} a_n$ 一定发散.

(D) 以上结论都不对.

(3) 曲面 $2x^2+y^2=z^2$ 是一个

(A) 双叶双曲面.　　(B) 椭圆锥.

(C) 单叶双曲面.　　(D) 椭圆抛物面.

(4) 若 $f(x,y)=\varphi(x+y)+\varphi(x-y)+\int_{x-y}^{x+y}\psi(t)dt$，其中 φ 和 ψ 二阶可导函数，则

(A) $\dfrac{\partial^2 f}{\partial x\partial y}=\dfrac{\partial^2 f}{\partial x^2}$.　　(B) $\dfrac{\partial^2 f}{\partial x\partial y}=\dfrac{\partial^2 f}{\partial y^2}$.

(C) $\dfrac{\partial^2 f}{\partial x^2}=-\dfrac{\partial^2 f}{\partial y^2}$.　　(D) $\dfrac{\partial^2 f}{\partial x^2}=\dfrac{\partial^2 f}{\partial y^2}$.

(5) $\lim\limits_{(x,y)\to(0,0)}(1+xy)^{\frac{1}{x^2+y^2}}=$

(A) 0 .　　(B) 1 .

(C) e .　　(D) 极限不存在.

二、　（15分）**填空题:**

(1) 若 $\mathbf{a},\mathbf{b},\mathbf{c}$ 为单位向量且满足 $\mathbf{a}+\mathbf{b}+\mathbf{c}=\mathbf{0}$，则 $\mathbf{a}\cdot\mathbf{b}+\mathbf{b}\cdot\mathbf{c}+\mathbf{c}\cdot\mathbf{a}=$ _______.

(2) 向量 $\mathbf{c}$ 垂直于向量 $\mathbf{a}=\langle\mathbf{1},\mathbf{2},\mathbf{1}\rangle$ 和向量 $\mathbf{b}=\langle\mathbf{-1},\mathbf{1},\mathbf{1}\rangle$，且满足 $\mathbf{c}\cdot(\mathbf{i}-\mathbf{2j}+\mathbf{k})=\mathbf{16}$，那么 $\mathbf{c}=$ _______ .

(3) 若 $\sum\limits_{n=2}^{\infty}\left(\tan\dfrac{1}{n}-k\ln\left(1-\dfrac{1}{n}\right)\right)$ 收敛，则 $k=$ _______ .

(4) 曲线 $y(x)=\sin x$ 的曲率 κ 的最大值为 _______.

(5) 若$(z+y)^x=xy$, 则$\frac{\partial z}{\partial x}(1,2)=$ _______.

三、　（10分）设 $a>0$, 方程$r=a(1+\cos\theta)$ 和 $r=a$ 分别表示一条心形线和一个圆周的方程.

(1) 求心形线所围成的区域中在圆外的部分的面积.

(2) 求心形线与圆相交的区域面积.

四、　（10分）设 $\mathbf{r}(t)=f(t)\,\mathbf{i}+g(t)\,\mathbf{j}+h(t)\mathbf{k}$, 其中

$$f(t)=\int_0^t\cos(x^2)\,dx,\quad g(t)=-t\cos t,\quad h(t)=\sum_{n=1}^{\infty}\frac{t^n}{n}.$$

计算$\mathbf{r}'(0)$.

五、　（10分）令$f(x,y)=\begin{cases} y\sin\frac{1}{x^2+y^2}, & (x,y)\neq(0,0); \\ 0, & (x,y)=(0,0). \end{cases}$

(1) $f(x,y)$在$(0,0)$处是否连续?

(2) 若$f_x(0,0)$和$f_y(0,0)$存在，求其值.

六、　（10分）若下列极限存在，求其极限值；若否，证明其极限不存在.

(1) $\lim\limits_{(x,y)\to(0,0)}\dfrac{xy}{\sqrt{x^2+y^2}}$.

(2) $\lim\limits_{(x,y)\to(0,0)}\dfrac{xy^3+2x^2y^4}{x^2+y^6}$.

七、　（10分）设幂级数 $f(x)=\sum\limits_{n=1}^{\infty}\dfrac{n+2}{n(n+1)}x^n$,

(1) 求幂级数 $f(x)$ 的收敛域.

(2) 求幂级数 $f(x)$ 的和函数.

八、　（10分）判断级数 $\sum\limits_{n=2}^{\infty}\dfrac{(-1)^{n+1}}{n^p(\ln n)^2}\,(p>0)$ 是否为绝对收敛、条件收敛或者发散，请给出你的理由.

九、　（10分）求 $\lim\limits_{n\to\infty}\left((n^2-n)e^{\frac{1}{n}}-\sqrt{n^4+1}\right)$.

A

1. (15pts) **Multiple Choice Questions:** (only one correct answer for each of the following questions.)

$z(x,y)$ 可导

(1) If f is differentiable, and $z = z(x,y)$ is determined by $f(x-az, y-bz) = 0$, then $a\frac{\partial z}{\partial x} + b\frac{\partial z}{\partial y} =$

(A) 1. (B) -1. (C) a (D) b.

$$\frac{\partial f(x-az, y-bz)}{\partial x} = \frac{\partial f(x-az, y-bz)}{\partial y} = 0$$

这里 $\frac{\partial f}{\partial x}$ 代替 $\frac{\partial f(x-az, y-bz)}{\partial x}$ 不是 $\frac{\partial f(x,y)}{\partial x}$

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial(x-az)}\left(\frac{\partial(x-az)}{\partial z}\cdot\frac{\partial z}{\partial x} + \frac{\partial(x-az)}{\partial x}\right) + \frac{\partial f}{\partial(y-bz)}\cdot\frac{\partial(y-bz)}{\partial z}\cdot\frac{\partial z}{\partial x}$$

$$= \frac{\partial f}{\partial(x-az)}\left(-a\frac{\partial z}{\partial x} + 1\right) + \frac{\partial f}{\partial(y-bz)}\left(-b\frac{\partial z}{\partial x}\right) = 0$$

$$\frac{\partial f}{\partial y} = \frac{\partial f}{\partial(y-bz)}\left(\frac{\partial(y-bz)}{\partial z}\cdot\frac{\partial z}{\partial y} + \frac{\partial(y-bz)}{\partial y}\right) + \frac{\partial f}{\partial(x-az)}\left(\frac{\partial(x-az)}{\partial z}\cdot\frac{\partial z}{\partial y}\right)$$

$$= \frac{\partial f}{\partial(y-bz)}\left(-b\frac{\partial z}{\partial y} + 1\right) + \frac{\partial f}{\partial(x-az)}\left(-a\frac{\partial z}{\partial y}\right) = 0$$

① $\frac{\partial f}{\partial(y-bz)}\left(-b\frac{\partial z}{\partial x}\right) = -\frac{\partial f}{\partial(x-az)}\left(-a\frac{\partial z}{\partial x} + 1\right)$

② $\frac{\partial f}{\partial(x-az)}\left(-a\frac{\partial z}{\partial y}\right) = -\frac{\partial f}{\partial(y-bz)}\left(-b\frac{\partial z}{\partial y} + 1\right)$

①×② $ab\frac{\partial z}{\partial x}\frac{\partial z}{\partial y}\quad\frac{\partial f}{\partial(x-az)}\frac{\partial f}{\partial(y-bz)} = \left(-a\frac{\partial z}{\partial x} + 1\right)\left(-b\frac{\partial z}{\partial y} + 1\right)\frac{\partial f}{\partial(x-az)}\frac{\partial f}{\partial(y-bz)}$

$$ab\frac{\partial z}{\partial x}\frac{\partial z}{\partial y} = \left(-a\frac{\partial z}{\partial x} + 1\right)\left(-b\frac{\partial z}{\partial y} + 1\right)$$

$$ab\frac{\partial z}{\partial x}\frac{\partial z}{\partial y} = ab\frac{\partial z}{\partial x}\frac{\partial z}{\partial y} - a\frac{\partial z}{\partial y} - b\frac{\partial z}{\partial y} + 1$$

$$\Rightarrow a\frac{\partial z}{\partial x} + b\frac{\partial z}{\partial y} = 1$$

C (2) Let $a_n > 0$ for all n. Which of the following statements must be **true** ?

(A) If $\lim_{n\to\infty} na_n = 0$, then the series $\sum_{n=1}^{\infty} a_n$ converges.

(B) If $\lim_{n\to\infty} na_n = l$ and $l \neq 0$, then the series $\sum_{n=1}^{\infty} a_n$ converges.

(C) If $\lim_{n\to\infty} na_n = l$ and $l \neq 0$, then the series $\sum_{n=1}^{\infty} a_n$ diverges.

(D) None of the above statements is correct.

(A) $a_n = \frac{1}{n\ln n}$

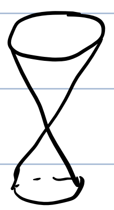
(C) $\lim_{n\to\infty} \frac{a_n}{\frac{1}{n}} = l$ $\quad \sum_{n=1}^{\infty} \frac{1}{n}$ diverges $\Rightarrow \sum_{n=1}^{\infty} a_n$ diverges

(B) $a_n = \frac{1}{n}$

B (3) Identify the surface of $2x^2 + y^2 = z^2$.

(A) Hyperboloid of two sheets. (B) Elliptical Cone.

(C) Hyperboloid of one sheet. (D) Elliptical paraboloid.



D (4) If $f(x,y) = \varphi(x+y) + \varphi(x-y) + \int_{x-y}^{x+y} \psi(t)dt$, where φ and ψ are twice differentiable functions, then

(A) $\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial x^2}$. (B) $\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y^2}$.

(C) $\frac{\partial^2 f}{\partial x^2} = -\frac{\partial^2 f}{\partial y^2}$. (D) $\frac{\partial^2 f}{\partial x^2} = \frac{\partial^2 f}{\partial y^2}$.

$$\frac{\partial f}{\partial x} = \frac{\partial \varphi(x+y)}{\partial(x+y)} + \frac{\partial \varphi(x-y)}{\partial(x-y)} + \psi(x+y) - \psi(x-y)$$

$$\frac{\partial f}{\partial y} = \frac{\partial \varphi(x+y)}{\partial(x+y)} - \frac{\partial \varphi(x-y)}{\partial(x-y)} + \psi(x+y) + \psi(x-y)$$

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 \varphi(x+y)}{\partial (x+y)^2} - \frac{\partial^2 \varphi(x-y)}{\partial (x-y)^2} + \frac{\partial \psi(x+y)}{\partial (x+y)} + \frac{\partial \psi(x-y)}{\partial (x-y)}$$

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial^2 \varphi(x+y)}{\partial (x+y)^2} + \frac{\partial^2 \varphi(x-y)}{\partial (x-y)^2} + \frac{\partial \psi(x+y)}{\partial (x+y)} - \frac{\partial \psi(x-y)}{\partial (x-y)}$$

$$\frac{\partial^2 f}{\partial y^2} = \frac{\partial^2 \varphi(x+y)}{\partial (x+y)^2} + \frac{\partial^2 \varphi(x-y)}{\partial (x-y)^2} + \frac{\partial \psi(x+y)}{\partial (x+y)} - \frac{\partial \psi(x-y)}{\partial (x-y)}$$

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial^2 f}{\partial y^2}$$

D (5) $\lim_{(x,y)\to(0,0)} (1+xy)^{\frac{1}{x^2+y^2}} =$

(A) 0 . (B) 1 .

(C) e . (D) does not exist.

$$\lim_{(x,kx)\to(0,0)} e^{\frac{\ln(1+kx^2)}{(1+k^2)x^2}} = e^{\frac{k}{1+k^2}}$$

2. (15 pts) Fill in the blanks.

(1) If $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are unit vectors and $\mathbf{a}+\mathbf{b}+\mathbf{c}=\mathbf{0}$, then $\mathbf{a}\cdot\mathbf{b}+\mathbf{b}\cdot\mathbf{c}+\mathbf{c}\cdot\mathbf{a} =$ $-\frac{3}{2}$.

$$(a+b+c)(a+b+c) = |a|^2+|b|^2+|c|^2+2a\cdot b+2b\cdot c+2c\cdot a = 0$$

$$a\cdot b+b\cdot c+c\cdot a = -\frac{3}{2}$$

(2) If the vector $\mathbf{c}$ is perpendicular to $\mathbf{a} = \langle \mathbf{1}, \mathbf{2}, \mathbf{1}\rangle$ and $\mathbf{b} = \langle -\mathbf{1}, \mathbf{1}, \mathbf{1}\rangle$ and $\mathbf{c}\cdot(\mathbf{i}-\mathbf{2j}+\mathbf{k}) = \mathbf{16}$, then $\mathbf{c} =$ $2i-4j+6k$

$$c' = a\times b = \begin{vmatrix} i & j & k \\ 1 & 2 & 1 \\ -1 & 1 & 1 \end{vmatrix} = i-2j+3k$$

$c = dc'$

$dc'\cdot(i-2j+k) = 16$

$d(1+4+3) = 16 \quad d=2$

$c = 2c'$

$$\tan x = x+\frac{x^3}{3}+\frac{2x^5}{15}+\cdots$$

(3) If $\sum_{n=2}^{\infty}\left(\tan\frac{1}{n} - k\ln\left(1-\frac{1}{n}\right)\right)$ converges, then $k =$ -1 .

$$\tan\frac{1}{n} = \frac{1}{n}+\frac{1}{3n^3}+o\left(\frac{1}{n^3}\right)$$

$$\ln\left(1-\frac{1}{n}\right) = -\frac{1}{n}-\frac{1}{n^2}+o\left(\frac{1}{n^2}\right)$$

$$\frac{1}{n}+\frac{k}{n} = 0 \qquad k=-1$$

(4) The maximum curvature κ of function $y(x) = \sin x$ is $\underline{\ 1\ }$.

$y'(x) = \cos x \quad y''(x) = -\sin x$

$$k(x) = \frac{|\sin x|}{(1+\cos^2 x)^{\frac{3}{2}}}$$

$k_{max} = 1$

(5) If $(z+y)^x = xy$, then $\frac{\partial z}{\partial x}(1,2) = \underline{2-2\ln 2}$.

$(z+y)^x = xy$

$e^{x\ln(z+y)} = xy$

$e^{x\ln(z+y)}\left(\ln(z+y) + \frac{x}{z+y}\frac{\partial z}{\partial x}\right) = y$

$(z+y)^x \ln(z+y) + (z+y)^{x-1}\frac{\partial z}{\partial x} = y$

$(z+y)^x\big|_{(1,2)} = xy\big|_{(1,2)}$

$(z+2)^1 = 2 \quad z=0$

Substitute $\begin{cases} x=1 \\ y=2 \\ z=0 \end{cases}$ $2\ln 2 + 2^0 \frac{\partial z}{\partial x} = 2$

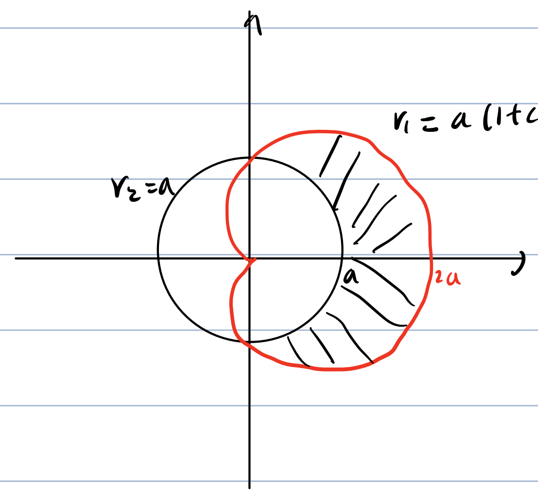
$$\frac{\partial z}{\partial x} = 2 - 2\ln 2$$

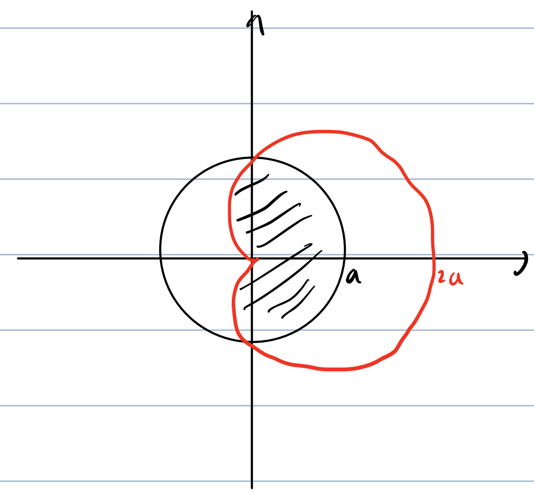
3. (10 pts) Given a cardioid $r = a(1+\cos\theta), a > 0$ and a circle $r = a$.

(1) Find the area of the region that lies inside the cardioid and outside the circle.

(2) Find the area of the region that lies inside the cardioid and inside the circle.

(1)





$$S = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{2}(r_1^2 - r_2^2)\,d\theta$$

$$S = \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \frac{1}{2}r_1^2\,d\theta + \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{2}r_2^2\,d\theta$$

$$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{2}\left(a^2(1+\cos\theta)^2 - a^2\right) d\theta$$

$$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(a^2\cos\theta + \frac{1}{2}a^2\cos^2\theta\right) d\theta$$

$$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(a^2\cos\theta + a^2\frac{1+\cos 2\theta}{4}\right) d\theta$$

$$= a^2\sin\theta + \frac{a^2}{4}\theta + \frac{a^2}{8}\sin 2\theta \Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}}$$

$$= 2a^2 + \frac{\pi a^2}{4}$$

$$= \frac{1}{2}\int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} a^2(1+\cos\theta)^2 d\theta + \frac{1}{2}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} a^2 d\theta$$

$$= \frac{1}{2}\int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} a^2(1+2\cos\theta+\cos^2\theta)\, d\theta + \frac{1}{2}\pi a^2$$

$$= \frac{a^2}{2}\int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \left(1+2\cos\theta+\frac{1+\cos 2\theta}{2}\right) d\theta + \frac{1}{2}\pi a^2$$

$$= \frac{a^2}{2}\left(\theta + 2\sin\theta + \frac{\theta}{2} + \frac{\sin 2\theta}{4}\right)\Big|_{\frac{\pi}{2}}^{\frac{3\pi}{2}} + \frac{1}{2}\pi a^2$$

$$= \frac{a^2}{2}\left(\pi - 4 + \frac{\pi}{2}\right) + \frac{1}{2}\pi a^2$$

$$= \frac{5\pi a^2}{4} - 2a^2$$

4. (10 pts) Assume $\mathbf{r}(t) = f(t)\,\mathbf{i} + g(t)\,\mathbf{j} + h(t)\mathbf{k}$, where

$$f(t) = \int_0^t \cos(x^2)\,dx, \quad g(t) = -t\cos t, \quad h(t) = \sum_{n=1}^{\infty} \frac{t^n}{n}.$$

Calculate $\mathbf{r}'(0)$.

$f'(t) = \cos t^2 \qquad g'(t) = -\cos t + t\sin t$

$\lim_{n\to\infty} \sqrt[n]{\left|\frac{1}{n}t^n\right|} = |t|$

$|t|<1 \qquad \sum_{n=1}^{\infty} \frac{t^n}{n}$ converges absolutely.

By Term-by-Term Differentiation Theorem:

$$h'(t) = \sum_{n=1}^{\infty} t^{n-1} = \frac{1}{1-t}$$

$$r'(t) = \cos t^2 i + (t\sin t - \cos t) j + \frac{1}{1-t} k$$

$$r'(0) = i - j + k.$$

6. (10 pts) Find the limit, if it exists, or show that the limit does not exist.

(1) $\lim_{(x,y)\to(0,0)} \dfrac{xy}{\sqrt{x^2+y^2}}$.

(2) $\lim_{(x,y)\to(0,0)} \dfrac{xy^3+2x^2y^4}{x^2+y^6}$.

(1) It exists.

$$\left|\frac{xy}{\sqrt{x^2+y^2}}\right| \leq \frac{\frac{1}{2}(x^2+y^2)}{\sqrt{x^2+y^2}} = \frac{1}{2}\sqrt{x^2+y^2}$$

$$\lim_{(x,y)\to(0,0)} \frac{1}{2}\sqrt{x^2+y^2} = 0$$

$$\lim_{(x,y)\to(0,0)} \frac{xy}{\sqrt{x^2+y^2}} = \lim_{(x,y)\to(0,0)} \left|\frac{xy}{\sqrt{x^2+y^2}}\right| = 0$$

(2) It does not exists.

Let $y = k\sqrt[3]{x}$

$$\lim_{(x,k\sqrt[3]{x})\to(0,0)} \frac{k^3x^2+2x^3k^4\sqrt[3]{x}}{x^2+k^6x^2} = \lim_{x\to 0} \frac{k^3+2k^4x\sqrt[3]{x}}{1+k^6} = \frac{k^3}{1+k^6}$$

The Limit is dependent on k.

So the formal limit does not exists.

8. (10 pts) Determine if the series, $\sum_{n=2}^{\infty} \dfrac{(-1)^{n+1}}{n^p(\ln n)^2}$ $(p > 0)$, converges absolutely, or converges conditionally, or diverges. Give reasons for your answer.

$p > 1$ Consider $\sum_{n=2}^{\infty} \frac{1}{n^p(\ln n)^2}$

$$\lim_{n\to\infty} \frac{\frac{1}{n^p(\ln n)^2}}{\frac{1}{n^p}} = \lim_{n\to\infty} \frac{1}{(\ln n)^2} = 0 \qquad \sum_{n=2}^{\infty} \frac{1}{n^p} \text{ converges}$$

By Limit Comparison Test, $\sum_{n=2}^{\infty} \frac{1}{n^p(\ln n)^2}$ converges

So $\sum_{n=2}^{\infty} \frac{1}{n^p(\ln n)^2}$ converges absolutely,

$0 < p \leq 1$ $\quad \lim_{n\to\infty} \frac{1}{n^p(\ln n)^2} = 0$

$$\forall n > 2 \quad 0 < \frac{1}{(n+1)^p(\ln(n+1))^2} \leq \frac{1}{n^p(\ln n)^2}$$

By Alternative Series Test, $\sum_{n=2}^{\infty} \frac{(-1)^{n+1}}{n^p(\ln n)^2}$ converges

$P=1$ Consider $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$

$$\int_2^{\infty} \frac{1}{x(\ln x)^2}\,dx \text{ and } \sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2} \text{ both converges or diverges.}$$

$$\int_2^{\infty} \frac{1}{x(\ln x)^2}\,dx = \int_2^{\infty} \frac{1}{(\ln x)^2}\,d\ln x = -\frac{1}{\ln x}\Big|_2^{\infty} = \frac{1}{\ln 2}$$

$\propto P<1$ $\lim_{n\to\infty} \frac{\frac{1}{n^p(\ln n)^2}}{\frac{1}{n}} = \lim_{n\to\infty} \frac{n^{1-p}}{(\ln n)^2} = \lim_{n\to\infty} \frac{(1-p)n^{1-p-1}}{2(\ln n)(\frac{1}{n})} = \lim_{n\to\infty} \frac{(1-p)n^{1-p}}{2\ln n} = \infty$

$\sum_{n=2}^{\infty} \frac{1}{n}$ diverges

By Limit Comparsion Test, $\sum_{n=2}^{\infty} \frac{1}{n^p(\ln n)^2}$ diverges

So $0<P<1$ $\sum_{n=2}^{\infty} \frac{(-1)^{n+1}}{n^p(\ln n)^2}$ converges conditionaly

$p \geq 1$ $\sum_{n=2}^{\infty} \frac{(-1)^{n+1}}{n^p(\ln n)^2}$ converges absolutely.

9. (10 pts) Find $\lim_{n\to\infty}\left((n^2-n)e^{\frac{1}{n}}-\sqrt{n^4+1}\right)$.

$\lim_{n\to\infty} o(1) = 0$

$$e^{\frac{1}{n}} = 1+\frac{1}{n}+\frac{1}{2n^2}+o\left(\frac{1}{n^2}\right)$$

$$\lim_{n\to\infty}\left((n^2-n)e^{\frac{1}{n}} - n^2\left(1+\frac{1}{n^4}\right)^{\frac{1}{2}}\right.$$

$$\lim_{n\to\infty}\left((n^2-n)\left(1+\frac{1}{n}+\frac{1}{2n^2}+o\left(\frac{1}{n^2}\right)\right)-n^2\left(1+\begin{bmatrix}\frac{1}{2}\\1\end{bmatrix}\frac{1}{n^4}+o\left(\frac{1}{n^4}\right)\right)\right)$$

$$=\lim_{n\to\infty}\left(n^2-n+n-1+\frac{1}{2}-\frac{1}{2n}+o(1)\right)-n^2\left(1+\frac{1}{2n^4}\right)+o\left(\frac{1}{n^2}\right)$$

$$=\lim_{n\to\infty}\left(n^2-\frac{1}{2}-\frac{1}{2n}\right)-n^2-\frac{1}{2n^2}+o(1)$$

$$=\lim_{n\to\infty}\left(-\frac{1}{2}-\frac{1}{2n}-\frac{1}{2n^2}+o(1)\right)$$

$$=-\frac{1}{2}$$